Design s Analysis of Algorithm

CSE211

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Week-7

CH.SC.U4CSE24136

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Q1)

Write a C program to implement Huffman coding that takes a string as input from the user, calculates the frequency of each character, and generates Huffman codes for all unique characters present in the string. Display the character, its frequency, and the corresponding Huffman code in a tabular format.

Consider the following input string representing a dataset label:

"DATAANALYTICSANDINTELLIGENCELABORATORY"

Code:

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_TREE_HT 100
#define MAX_CHAR 256

struct MinHeapNode {
    char data;
    unsigned freq;
    struct MinHeapNode *left, *right;
};

struct MinHeap {
    unsigned size;
    unsigned capacity;
    struct MinHeapNode** array;
};

struct MinHeapNode* newNode(char data, unsigned freq) {
    struct MinHeapNode* temp =
        (struct MinHeapNode*)malloc(sizeof(struct MinHeapNode));
    temp->left = temp->right = NULL;
    temp->data = data;
    temp->freq = freq;
    return temp;
}

struct MinHeap* createMinHeap(unsigned capacity) {
    struct MinHeap* minHeap =
        (struct MinHeap*)malloc(sizeof(struct MinHeap));
    minHeap->size = 0;
    minHeap->capacity = capacity;
    minHeap->array =
        (struct MinHeapNode**)malloc(minHeap->capacity *
                                     sizeof(struct MinHeapNode*));
    return minHeap;
}

void swapMinHeapNode(struct MinHeapNode** a,
                     struct MinHeapNode** b) {
    struct MinHeapNode* t = *a;
    *a = *b;
    *b = t;
}

void minHeapify(struct MinHeap* minHeap, int idx) {
    int smallest = idx;
    int left = 2 * idx + 1;
    int right = 2 * idx + 2;

    if (left < minHeap->size &&
        minHeap->array[left]->freq <
        minHeap->array[smallest]->freq)
        smallest = left;
```

```
    if (right < minHeap->size &&
        minHeap->array[right]->freq <
        minHeap->array[smallest]->freq)
        smallest = right;

    if (smallest != idx) {
        swapMinHeapNode(&minHeap->array[smallest],
                        &minHeap->array[idx]);
        minHeapify(minHeap, smallest);
    }
}

struct MinHeapNode* extractMin(struct MinHeap* minHeap) {
    struct MinHeapNode* temp = minHeap->array[0];
    minHeap->array[0] =
        minHeap->array[minHeap->size - 1];

    --minHeap->size;
    minHeapify(minHeap, 0);

    return temp;
}

void insertMinHeap(struct MinHeap* minHeap,
                   struct MinHeapNode* minHeapNode) {

    ++minHeap->size;
    int i = minHeap->size - 1;

    while (i && minHeapNode->freq <
           minHeap->array[(i - 1) / 2]->freq) {

        minHeap->array[i] =
            minHeap->array[(i - 1) / 2];
        i = (i - 1) / 2;
    }

    minHeap->array[i] = minHeapNode;
}

void buildMinHeap(struct MinHeap* minHeap) {
    int n = minHeap->size - 1;
    for (int i = (n - 1) / 2; i >= 0; --i)
        minHeapify(minHeap, i);
}

int isLeaf(struct MinHeapNode* root) {
    return !(root->left) && !(root->right);
}

void printCodes(struct MinHeapNode* root,
                int arr[], int top) {

    if (root->left) {
        arr[top] = 0;
        printCodes(root->left, arr, top + 1);
    }
```

```c
    if (root->right) {
        arr[top] = 1;
        printCodes(root->right, arr, top + 1);
    }

    if (isLeaf(root)) {
        printf("%c | %d | ", root->data, root->freq);
        for (int i = 0; i < top; ++i)
            printf("%d", arr[i]);
        printf("\n");
    }
}

void calculateSpaceSaving(struct MinHeapNode* root,
                          int arr[], int top,
                          int* totalOriginalBits,
                          int* totalCompressedBits) {

    if (root->left) {
        arr[top] = 0;
        calculateSpaceSaving(root->left, arr, top + 1,
                             totalOriginalBits, totalCompressedBits);
    }

    if (root->right) {
        arr[top] = 1;
        calculateSpaceSaving(root->right, arr, top + 1,
                             totalOriginalBits, totalCompressedBits);
    }

    if (isLeaf(root)) {
        *totalOriginalBits += root->freq * 8;
        *totalCompressedBits += root->freq * top;
    }
}

void HuffmanCodes(char data[], int freq[], int size) {
    struct MinHeapNode *left, *right, *top;
    struct MinHeap* minHeap = createMinHeap(size);

    for (int i = 0; i < size; ++i)
        minHeap->array[i] = newNode(data[i], freq[i]);

    minHeap->size = size;
    buildMinHeap(minHeap);

    while (minHeap->size != 1) {
        left = extractMin(minHeap);
        right = extractMin(minHeap);

        top = newNode('$', left->freq + right->freq);
        top->left = left;
        top->right = right;

        insertMinHeap(minHeap, top);
    }

    int arr[MAX_TREE_HT], topIdx = 0;
```

```c
    printf("\nChar | Freq | Huffman Code\n");
    printf("----------------------------\n");

    struct MinHeapNode* root = extractMin(minHeap);
    printCodes(root, arr, topIdx);

    int totalOriginalBits = 0, totalCompressedBits = 0;

    calculateSpaceSaving(root, arr, 0,
                         &totalOriginalBits,
                         &totalCompressedBits);

    printf("\nSpace Saving Analysis:\n");
    printf("----------------------\n");
    printf("Original size: %d bits\n", totalOriginalBits);
    printf("Compressed size: %d bits\n", totalCompressedBits);
    printf("Space saved: %d bits (%.2f%%)\n",
           totalOriginalBits - totalCompressedBits,
           ((float)(totalOriginalBits - totalCompressedBits) /
            totalOriginalBits) * 100);
}

int main() {
    printf("CH.SC.U4CSE24136");
    char str[1000];
    int freq[MAX_CHAR] = {0};

    printf("Enter the string: ");
    scanf("%s", str);

    for (int i = 0; str[i] != '\0'; i++) {
        freq[(unsigned char)str[i]]++;
    }

    char unique_chars[MAX_CHAR];
    int unique_freqs[MAX_CHAR];
    int n = 0;

    for (int i = 0; i < MAX_CHAR; i++) {
        if (freq[i] > 0) {
            unique_chars[n] = (char)i;
            unique_freqs[n] = freq[i];
            n++;
        }
    }

    HuffmanCodes(unique_chars, unique_freqs, n);

    return 0;
}
```

Output:

```
CH.SC.U4CSE24136
Enter the string: DATAANALYTICSANDINTELLIGENCELABORATORY

Char | Freq | Huffman Code
----------------------------
N : 4 | 000
L : 4 | 001
O : 2 | 0100
R : 2 | 0101
C : 2 | 0110
B : 1 | 01110
G : 1 | 01111
T : 4 | 100
Y : 2 | 1010
S : 1 | 10110
D : 2 | 10111
E : 3 | 1100
I : 3 | 1101
A : 7 | 111

Space Saving Analysis:
----------------------
Original size (8 bits per character): 304 bits
Compressed size (Huffman coding): 138 bits
Space saved: 166 bits (54.61%)
```

Time Complexity: O(N+ M log M) : O(n + m log m) where N is the input string length and M is the number of unique characters - the frequency counting takes O(N) while building the Huffman tree with m nodes takes O(M log M) due to repeated extractMin and insert operations in the min-heap.

Space Complexity: O(N+M) : O(N + M) where N stores the input string and M stores the Huffman tree nodes and related data structures - since M is bounded by 256 (ASCII characters), this effectively becomes O(N) for practical purposes.